# Common two-factor authentication methods

**Using a verification email**

Stronger than just having a password but not considered as second factor by NIST

**Using mobile device**

- by using a push approval application (weak)
- by sending a one-time password through an SMS
- by using a one-time password application

**Using a hardware authentication device**

e.g Hardware Token/Fob, Smart Cards, USB Security Keys

# One-time password

**Registration:** The service provide users with unique secret keys (using a QR code usually) to store in their mobile application

**Authentication:** The device generates a one-time password based on the secret stored in the device

There are two standards both based on MAC

- HOTP (HMAC-based one-time password) based on a counter
- **TOTP** (Time-based one-time password)
  based on time quantum ~ usually 30 seconds